Entity relationship diagram: how to read it

The data of Lakbay Salita: game content, player data and runtime hand-over

Entity box



Header: the entity name and, in small type, the file or PlayerPrefs key that stores it.

PK = primary key (shaded rows): the attributes that identify one record.

FK = foreign key: refers to another entity.

Right column = data type (int, string, bool, enum; string[] is a list).

Dashed border = runtime-only data that is never saved (page 5).

Relationship ends (crow's foot notation)



Read a line from one entity to the other: 'one Topic has many WordleLevel records'.



Dashed line = soft link: a stored index or word that points to another entity; not enforced.

Pages

2 Complete ERD: all entities on one page

3 Game content (read-only JSON files)

4 Player data (saved on the computer)

5 Runtime hand-over between scenes (not saved)

6 Data dictionary: content entities

7 Data dictionary: player and runtime entities

Colour key

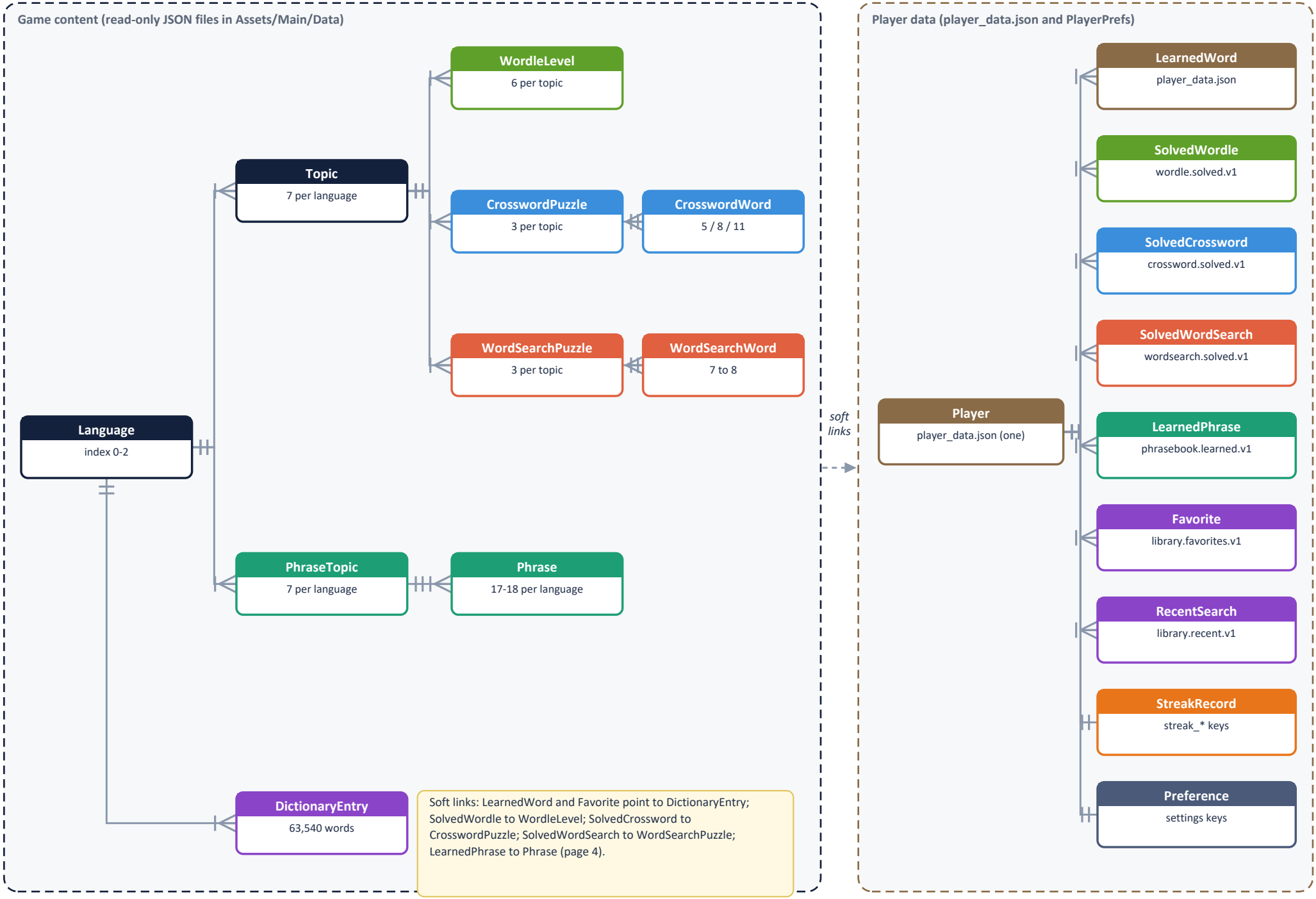
- Language and Topic
- Wordle
- Crossword
- Word Search
- Phrasebook
- Dictionary (Library)
- Player record
- Streak
- Preferences

Important:

- This is a logical model. The game has no database: content is read from JSON files, and the player's progress is kept in player_data.json and in PlayerPrefs lists.
- Keys such as 'language:category:level' are strings; nothing checks that they point to a record that still exists.

Complete ERD

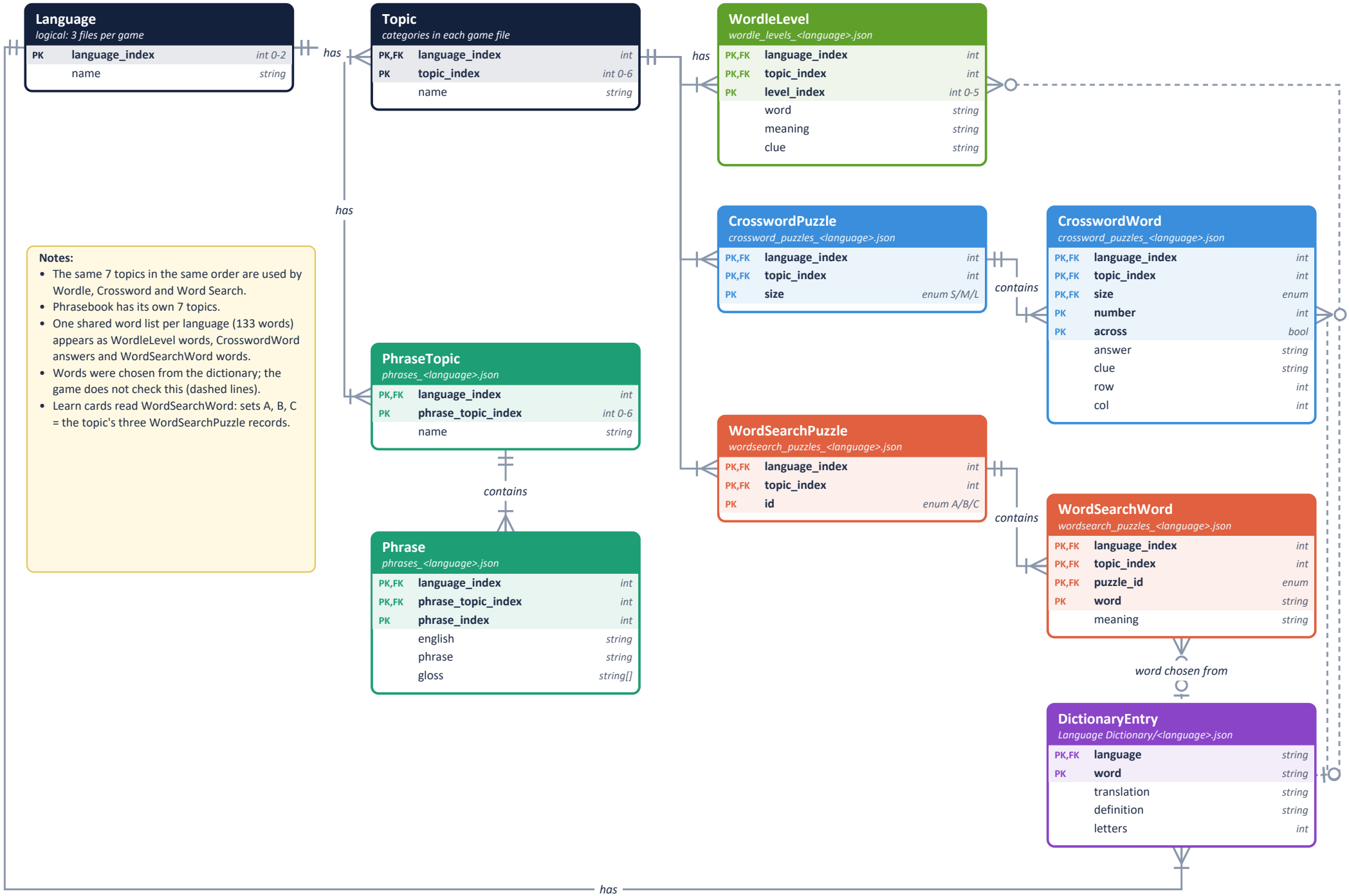
All entities of the game. Names only; attributes are on pages 3 to 5



Logical model: the game stores indexes and words in JSON files and PlayerPrefs; it has no database, so no foreign key is enforced.

Game content (read-only JSON)

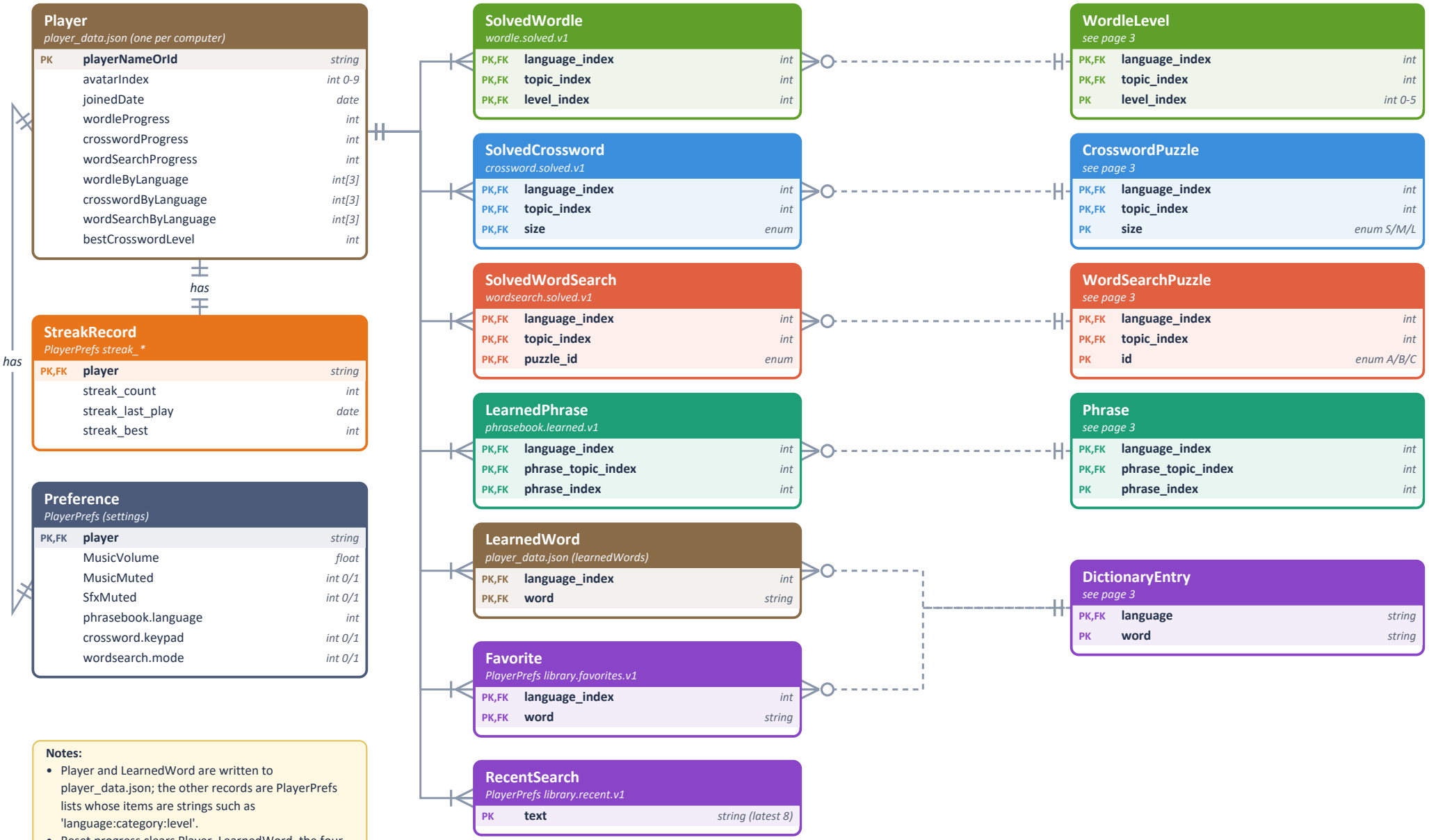
Everything the games read: topics, levels, puzzles, phrases and the dictionary



Logical model: the game stores indexes and words in JSON files and PlayerPrefs; it has no database, so no foreign key is enforced.

Player data (saved on the computer)

player_data.json and PlayerPrefs lists, with soft links to the game content



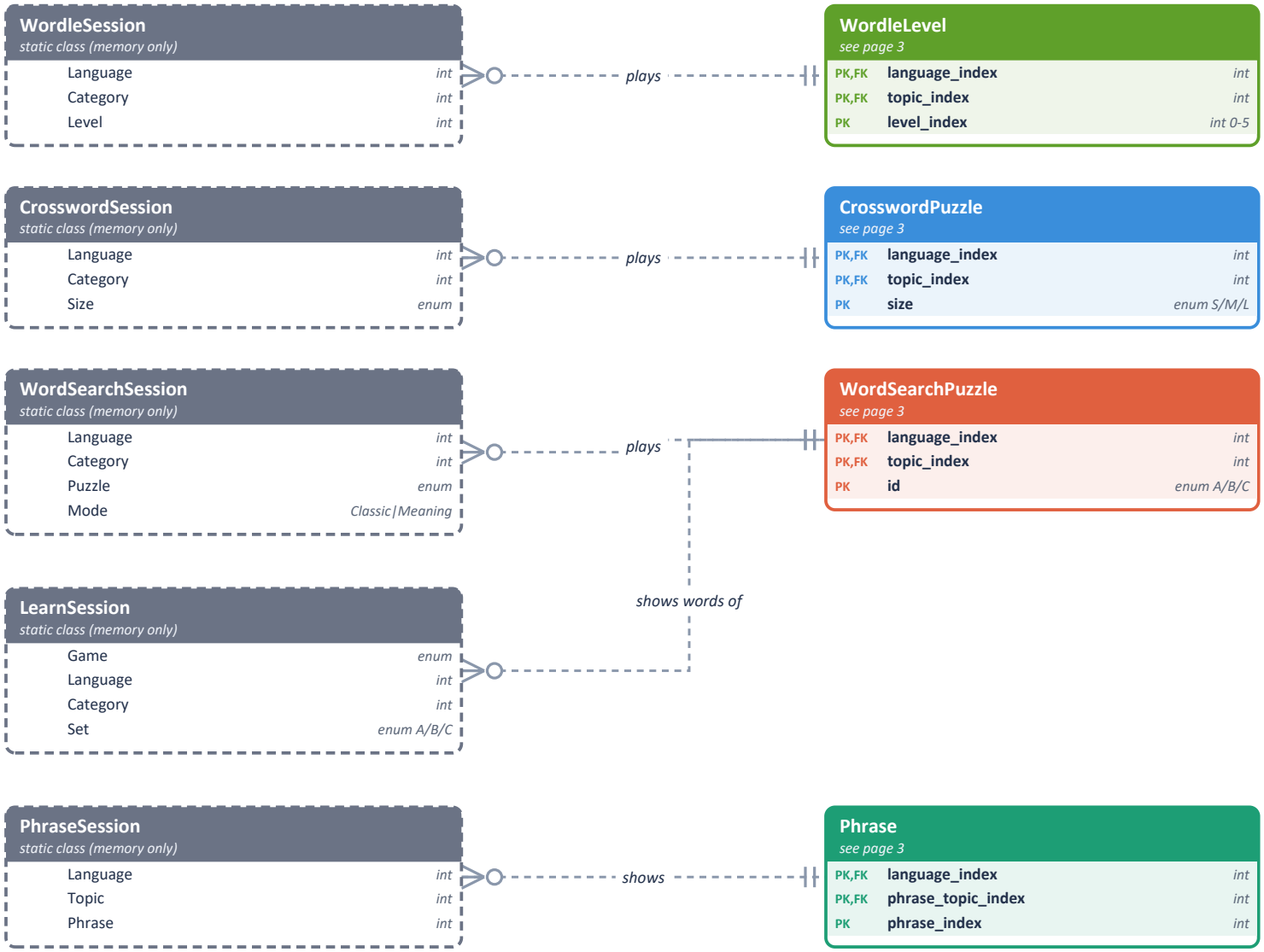
Notes:

- Player and LearnedWord are written to player_data.json; the other records are PlayerPrefs lists whose items are strings such as 'language:category:level'.
- Reset progress clears Player, LearnedWord, the four Solved / Learned lists and the StreakRecord.
- Favorite, RecentSearch and Preference are kept after a reset.
- A Player owns the records by being the only player on the computer; there is no player key in the stored strings.

Logical model: the game stores indexes and words in JSON files and PlayerPrefs; it has no database, so no foreign key is enforced.

Runtime hand-over between scenes (not saved)

Static Session classes: one scene fills them, the next scene reads them



How it works:

- A scene (a Start screen, the Language hub, a Learn button or a Phrasebook card) sets the fields and loads the next scene.
- The next scene reads them in Awake or Start, for example WordleCategoryBootstrap.
- Nothing here is saved: when the game closes the values are gone.
- LearnSession.Game says which game opened the Learn cards, so Start puzzle and Back return to that game.
- WordSearchSession.Mode is remembered in the Preference record (wordsearch.mode); the Phrasebook language in phrasebook.language.

Each field is an index or a letter that points into the content on page 3: Language = language_index, Category = topic_index, Level = level_index, Size = size, Puzzle = puzzle id, Topic = phrase_topic_index, Phrase = phrase_index.

Logical model: the game stores indexes and words in JSON files and PlayerPrefs; it has no database, so no foreign key is enforced.

Data dictionary: game content

Where each content entity lives and how it is identified

Entity	Stored in	Identifier (key)	Attributes	Count / notes
Language	logical (3 files per game, array order)	language_index 0-2	name (Cebuano, Ilonggo, Tagalog)	3 languages
Topic	categories list in each game JSON	language_index + topic_index	name	7 per language; same order in Wordle, Crossword, Word Search
WordleLevel	Data/Wordle/Levels/wordle_levels_<language>.json	language + topic + level_index	word (A-Z, 4-7 letters), meaning, clue	6 per topic, 42 per language, 126 in all
CrosswordPuzzle	Data/Crossword/Puzzles/crossword_puzzles_<language>.json	language + topic + size	size (Small, Medium, Large)	3 per topic, 21 per language, 63 in all
CrosswordWord	same file	puzzle + number + across	answer, clue, row, col	5 / 8 / 11 words per puzzle
WordSearchPuzzle	Data/Word Search/Puzzles/wordsearch_puzzles_<language>.json	language + topic + id (A, B, C)	id	3 per topic, 21 per language, 63 in all; grid is generated at run time
WordSearchWord	same file	puzzle + word	word (3-8 letters), meaning	7 to 8 per puzzle
PhraseTopic	Data/Phrasebook/phrases_<language>.json	language + phrase_topic_index	name	7 per language
Phrase	same file	language + topic + phrase_index	english, phrase, gloss[] (empty or one entry per word)	18 Cebuano, 17 Ilonggo, 18 Tagalog = 53
DictionaryEntry	Data/Language Dictionary/cebuano.json, hiligaynon.json, tagalog.json	language + word	translation, definition, letters	22,005 + 15,223 + 26,312 = 63,540

Logical model: the game stores indexes and words in JSON files and PlayerPrefs; it has no database, so no foreign key is enforced.

Data dictionary: player and runtime entities

What is saved, where, and what a reset clears

Entity	Stored in	Identifier (key)	Attributes	Count / notes
Player	player_data.json (Application.persistentDataPath)	one record per computer	playerNameOrId, avatarIndex 0-9, joinedDate, wordle / crossword / wordSearch progress, the same by language [3], bestCrosswordLevel	1; reset by Reset progress
LearnedWord	player_data.json, list learnedWords	language + word, stored as 'language:WORD'	(key only)	grows with play; reset by Reset progress
SolvedWordle	PlayerPrefs wordle.solved.v1	'language:category:level' joined with |	(key only)	up to 126
SolvedCrossword	PlayerPrefs crossword.solved.v1	'language:category:size'	(key only)	up to 63
SolvedWordSearch	PlayerPrefs wordsearch.solved.v1	'language:category:puzzle'	(key only)	up to 63
LearnedPhrase	PlayerPrefs phrasebook.learned.v1	'language:topic:phrase'	(key only)	up to 53
Favorite	PlayerPrefs library.favorites.v1	'language:WORD' joined with |	(key only)	any number; kept after Reset progress
RecentSearch	PlayerPrefs library.recent.v1	the search text	(text only)	latest 8, one per line; kept after Reset progress
StreakRecord	PlayerPrefs streak_count, streak_last_play, streak_best	one record	count, last play day (yyyy-MM-dd), best	1; reset by Reset progress
Preference	PlayerPrefs MusicVolume, MusicMuted, SfxMuted, phrasebook.language, crossword.keypad, wordsearch.mode	one record	volume 0-1; mute switches; last Phrasebook language; keypad switch; Word Search mode	1; kept after Reset progress
WordleSession	static class WordleSession (memory)	none	Language, Category, Level	runtime only
CrosswordSession	static class CrosswordSession (memory)	none	Language, Category, Size	runtime only
WordSearchSession	static class WordSearchSession (memory)	none	Language, Category, Puzzle, Mode	runtime only; Mode is also saved as wordsearch.mode
LearnSession	static class LearnSession (memory)	none	Game, Language, Category, Set	runtime only
PhraseSession	static class PhraseSession (memory)	none	Language, Topic, Phrase	runtime only; last language also saved as phrasebook.language

Logical model: the game stores indexes and words in JSON files and PlayerPrefs; it has no database, so no foreign key is enforced.